Università degli Studi di Messina | 2025/2026

An Optimized and Interpretable Deep Learning Framework for Brain Tumor Classification

Bachelor's Degree in Computer Science (L31)
Curriculum: Data Analysis

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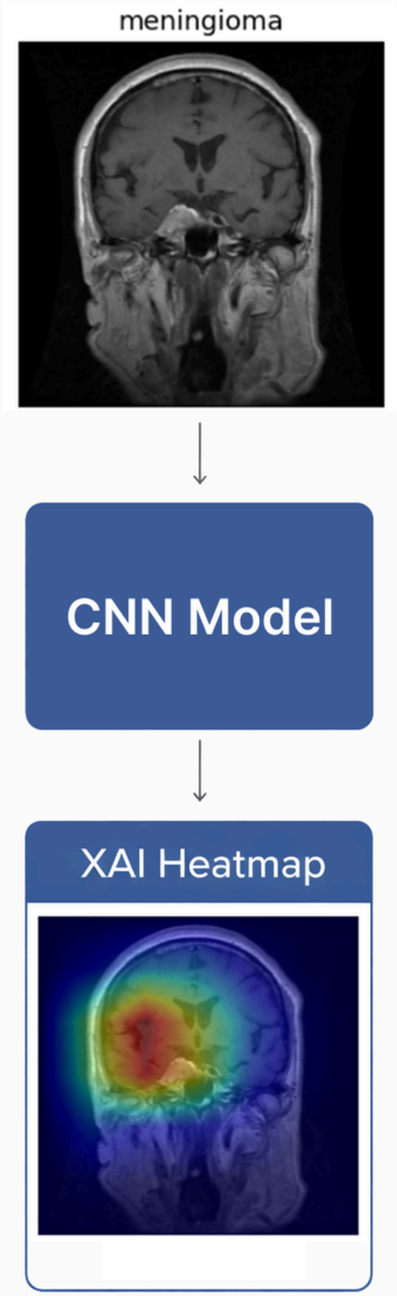
Background

- Brain tumors require early diagnosis
- MRI is the main imaging modality
- CNNs achieve high accuracy
- Interpretability is critical in medical AI



Objectives & Research Questions

Main Objective

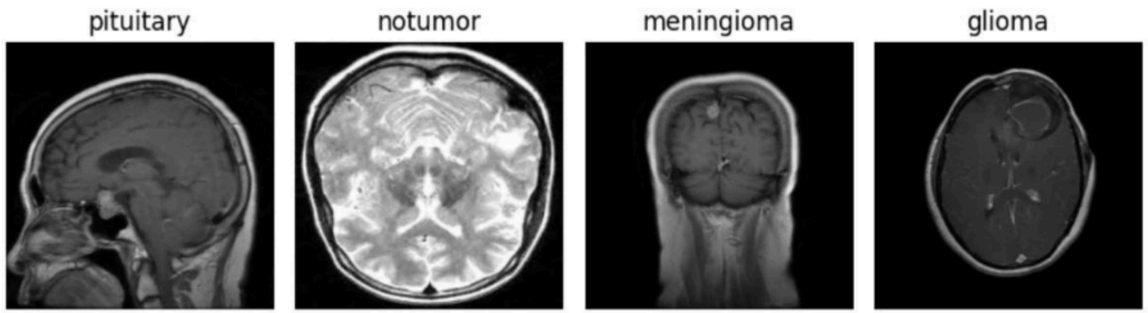
Design an interpretable EfficientNetV2-S framework for brain tumor MRI classification.

Specific Objectives

- Optimize EfficientNetV2-S
- Evaluate attention modules
- Compare XAI methods
- Analyze accuracy–interpretability trade-off

Research Questions

- Can EfficientNetV2-S be optimized without losing interpretability?
- Do attention modules improve performance?
- How does preprocessing affect generalization and XAI quality?



Dataset & Experimental Protocol

Dataset Overview

- Brain Tumor MRI dataset (Kaggle)
- 6,597 unique MRI slices (after cleaning)
- 4 classes: glioma, meningioma, pituitary, no tumor

Data Integrity

- 426 duplicates removed (SHA-1 hashing)

Experimental Protocol

- Stratified split: 80 / 10 / 10
- Train: 5,343 Validation: 594 Test: 660
- Input size: 224×224×3

Model Architecture

EfficientNetV2-S Backbone

- ImageNet pre-trained
- Compound scaling (depth, width, resolution)
- Fused-MBConv / MBConv blocks
- Built-in Squeeze-and-Excitation (SE) attention

Custom Classification Head

- Global Average Pooling
- Dense(256, ReLU)
- Dropout (0.5)
- Dense(4, Softmax)

Optimization (Final Configuration)

- Categorical Cross-Entropy
- AdamW with weight decay
- LR scheduling
- Early stopping

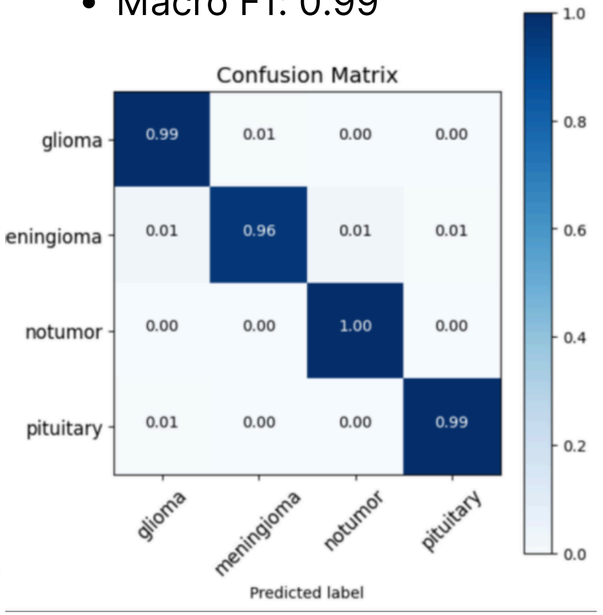
Experiment 1 — Baseline (M1)

Configuration

- Adam (lr=1e-3), batch=16, 50 epochs
- Standard augmentation

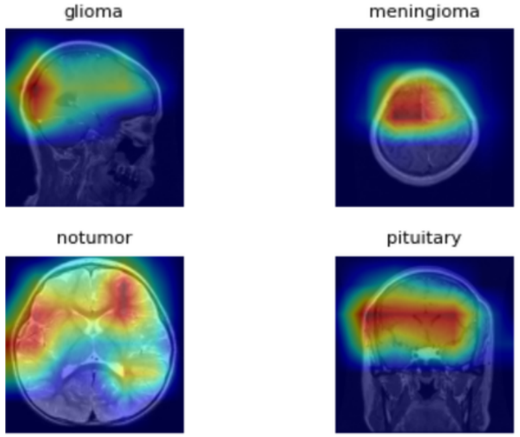
Results

- Accuracy: 98.77%
- Macro F1: 0.99



Interpretability

- Tumor-focused heatmaps
- Slightly diffuse activations



Observation

- Near state-of-the-art accuracy
- Further optimization required

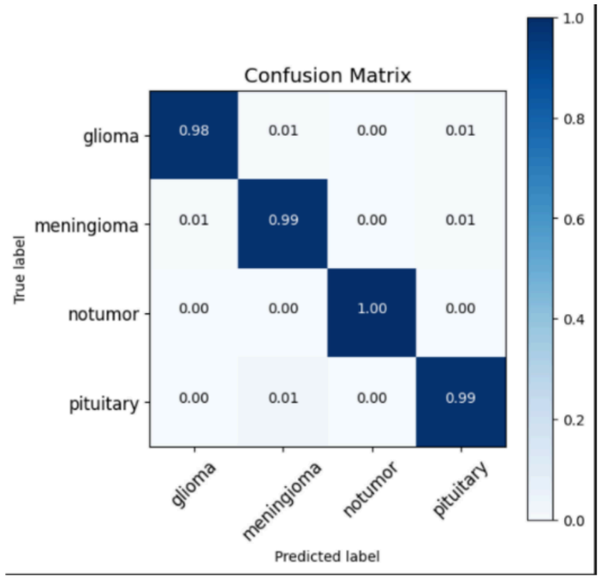
Experiment 2 — Training Optimization (M2)

Configuration

- AdamW (lr = 5e-4, wd = 1e-4)
- LR scheduling + early stopping
- Enhanced augmentation

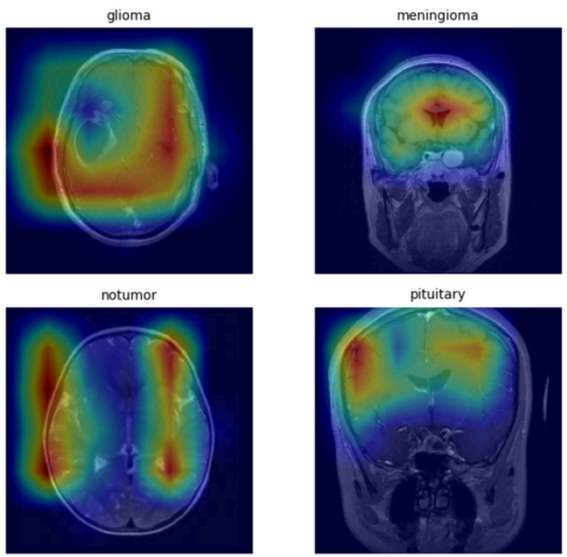
Results

- Accuracy: 98.95%
- Macro F1: 0.99
- More uniform confusion matrix



Interpretability

- Tumor-focused heatmaps
- Localization more stable (still slightly diffuse)



Observation

- Slight improvement in accuracy
- Better training stability
- More consistent class-wise predictions

Experiment 3 — Attention Modules (M3)

Motivation

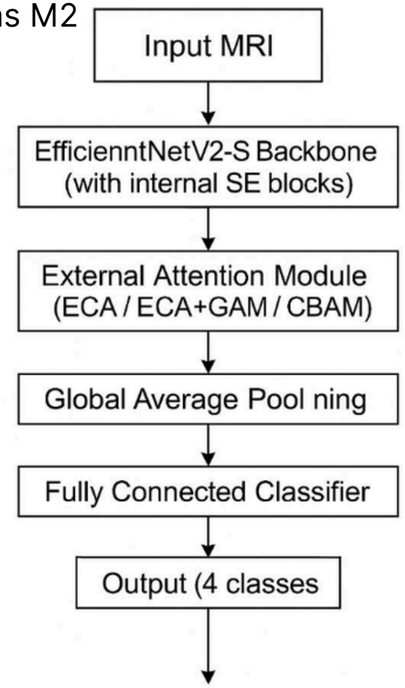
- EfficientNetV2-S already includes SE attention
- Test external attention modules

Tested modules

- ECA
- ECA + GAM
- CBAM

Configuration

- Attention inserted after backbone
- Same optimizer and training setup as M2



Results

Model Variant	Test Accuracy
ECA	0.9843
ECA + GAM	0.9800
CBAM	0.9740

- Accuracy decreased by ~1–1.5%
- Slight overfitting

Observation

- External attention did not improve performance
- Redundant with built-in SE blocks
- XAI analysis was not performed due to reduced model reliability.

Experiment 4 — Improved Preprocessing (M4)

Motivation

- Focus on data quality refinement

Changes

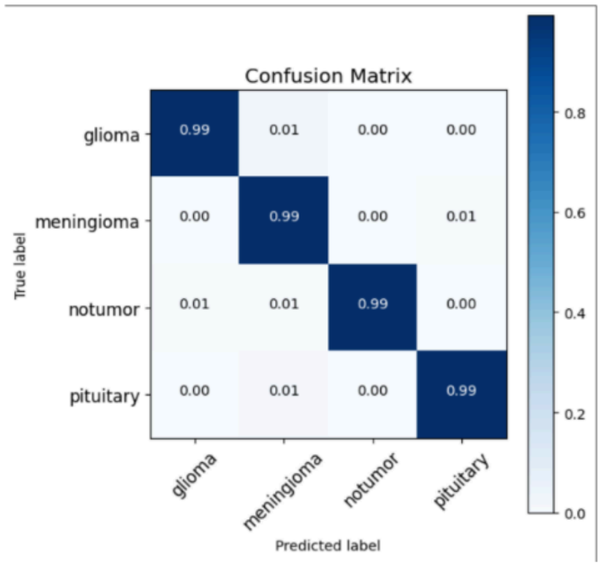
- 426 duplicates removed (SHA-1)
- Softer augmentation
- Increased dropout (0.5)

Results

- Accuracy: 98.94%
- More uniform confusion matrix
- More localized saliency maps

Key Insight

Data curation improved stability and interpretability



Experiment 5 — Label Smoothing (M5)

Configuration

- M4 + label smoothing (ε = 0.05)

Why Introduced?

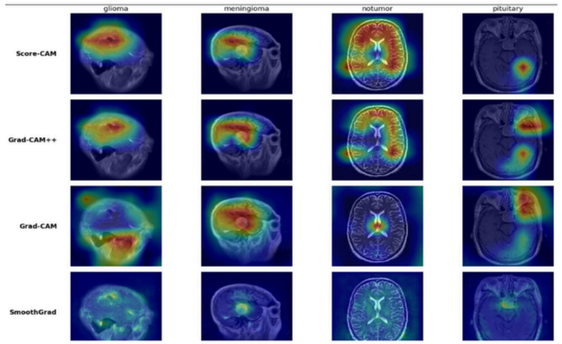
- Reduce overconfidence
- Smoother decision boundaries

Results

- Accuracy: 99.00% (highest)
- Macro F1-score: 0.99

Interpretability

- More diffuse Grad-CAM maps than M4
- Reduced localization precision
- Activations extend beyond tumor boundaries



Observation

- Slight accuracy improvement
- Localization became less precise
- M4 remains preferable for deployment

Why M4 Was Selected as Final Model

Model	Accuracy	Robustness	Interpretability	Verdict
M2 (Optimized Training)	98.95%	Moderate	Improved	Strong optimization baseline
M3 (Attention)	97.4–98.4%	Lower	Not evaluated	Redundant complexity
M4 (Improved Preprocessing)	98.94%	Strong	Sharp & stable XAI	Selected Model
M5 (Label Smoothing)	99.00%	Good	Diffuse heatmaps	Accuracy–XAI trade-off

Comparative Summary

- M5 achieved the highest accuracy but reduced localization sharpness.
- M2 improved training stability
- M4 provided the best balance of accuracy, robustness, and interpretability

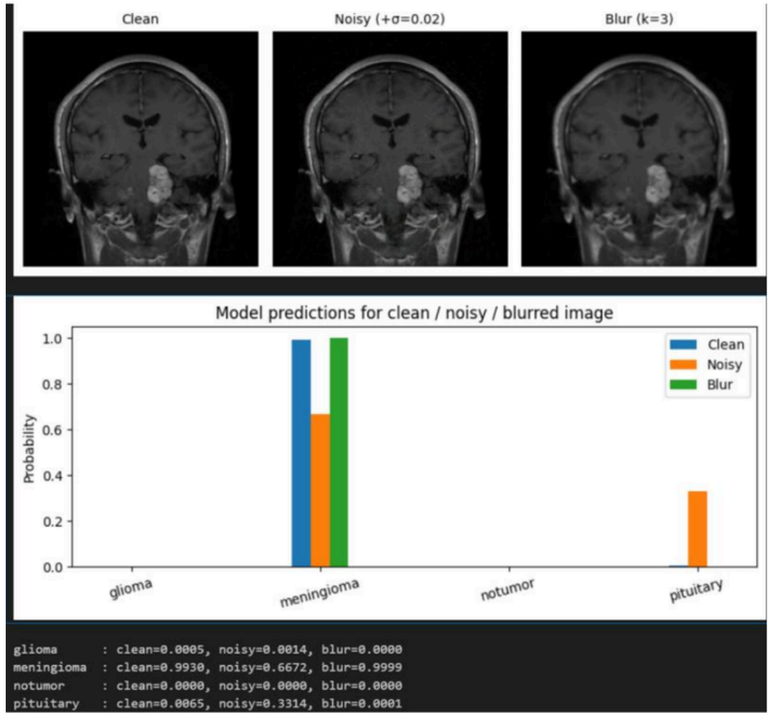
Final Decision Principle

M4 was selected because it balances performance, robustness, and interpretability — not accuracy alone.

Robustness Evaluation (Clinical Stress Test)

Stress Test Results

Condition	Accuracy
Clean	98.94%
Gaussian noise	98.03%
Contrast scaling	98.79%
Gamma correction	98.64%
Motion blur	93.79%
Resolution ↓	90.61%



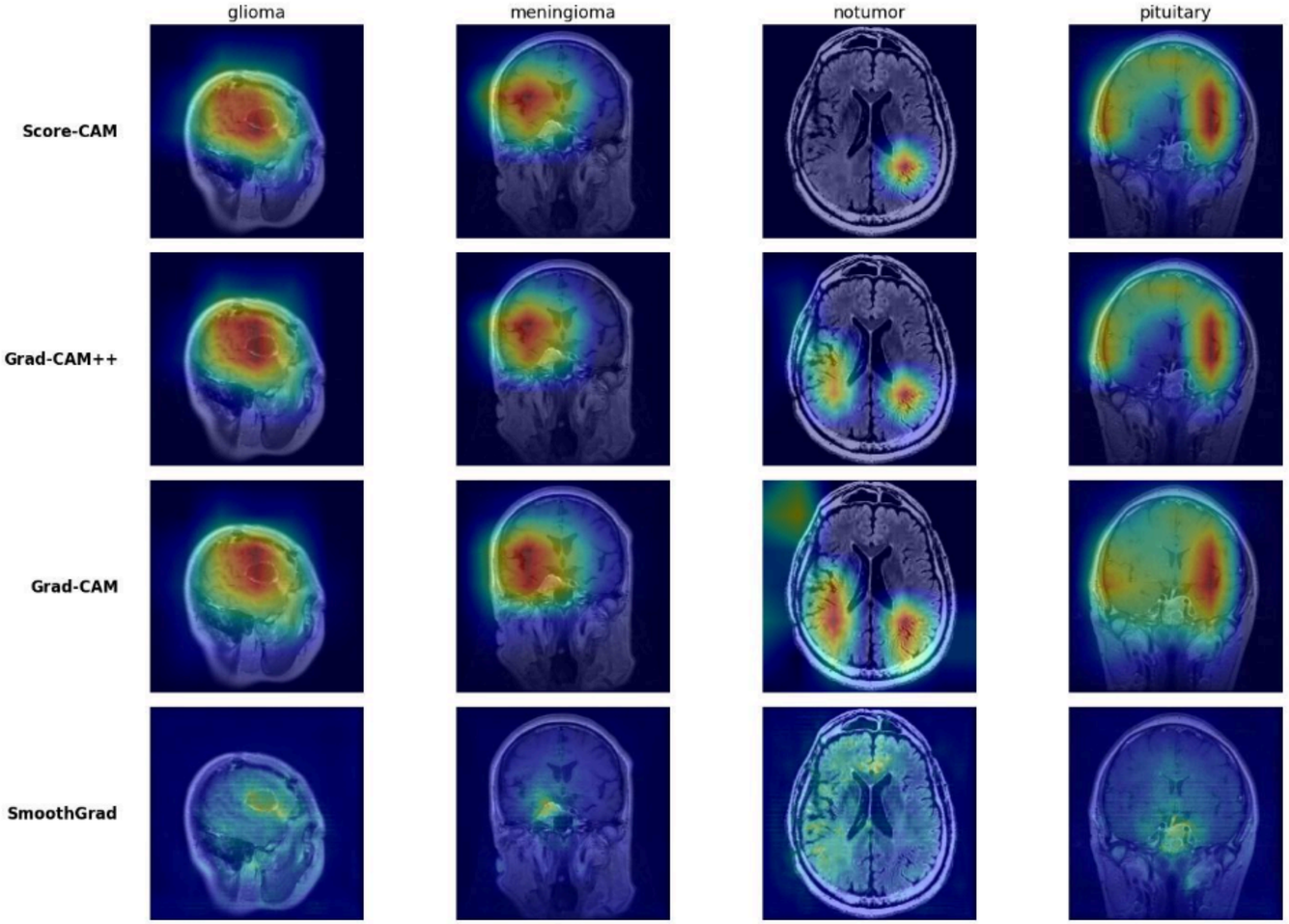
Key Findings

- Robust to noise and intensity changes
- Motion blur reduces performance
- Resolution loss has strongest impact

Clinical Insight

M4 is robust to MRI variability but sensitive to spatial detail loss.

Explainability Validation (Final Model – M4)



Methods Used

- Score-CAM
- Grad-CAM / Grad-CAM++
- SmoothGrad

Key Findings

- Consistent localization across XAI methods
- Activations aligned with tumor regions
- No spurious attention in no-tumor class

Conclusion

- Stable and anatomically meaningful localization

Contributions

Main contributions:

- Optimized EfficientNetV2-S framework
- Evaluation of attention mechanisms (ECA, GAM, CBAM) showing limited benefit for SE-based architectures
- Improved preprocessing pipeline enhancing robustness and localization stability
- Analysis of the accuracy–interpretability trade-off
- Multi-method XAI validation confirming consistent tumor localization

Conclusion

Key Outcomes

- Achieved ~98.9% reliable classification accuracy
- Maintained balanced per-class performance
- Demonstrated robustness under realistic clinical perturbations
- Validated interpretability using multiple XAI methods
- Identified a clear accuracy–interpretability trade-off

Final model selection was based on performance, robustness, and explainability — not accuracy alone.